

1(a)(i)	<i>independent:</i> mass / biomass, of yeast ; or concentration of, yeast / suspension / culture ; <i>dependent:</i> absorbance / colour (change / of indicator) ;	2
1(a)(ii)	<i>any two of:</i> volume of water (added) / volume of yeast suspension ; (same) indicator / TTC ; (same) temperature ; (same) time / period, of incubation ; colorimeter zeroed ;	2
1(a)(iii)	<i>one of:</i> <i>replace (live) yeast by:</i> boiled / dead / inactivated / AW, yeast (of the same mass) ; sterile / inert / glass beads / AW, material (of same mass) ;	1
1(a)(iv)	axes labelled $x =$, time / t , $y =$, absorbance / A_b ; line to show absorbance shows an increase ; ecf if axes inverted but shape (related to axes) correct = 1 ecf if axes wrong but shape correct (related to those axes) = 1	2

Question	Answer	Marks
1(b)	<i>any seven of:</i> 1 same / stated / known, mass / volume (suspension), of each yeast (added to separate flasks) ; 2 same / stated / known, concentration of, nutrient solution / glucose ; 3 same / stated / known, volume of, nutrient solution / glucose ; 4 <i>ref. to</i> method to maintain temperature ; 5 suitable temperature in range 15 °C–80 °C ; 6 <i>idea of</i> equilibration / bringing yeast suspension and nutrient solution, to temperature, before mixing ; 7 add TTC / redox indicator, to yeast / yeast and nutrient mixture ; 8 <i>time for recording absorbance</i> either record (absorbance) at, regular / stated, time intervals or leave for set time (if stated max = 1 hour) and record (absorbance) ; 9 <i>ref. to</i> method of maintaining homogeneity (of yeast) ; 10 <i>ref. to</i> method of maintaining oxygen concentration ; 11 use (at least) 3 replicates / repeats and find mean or identify / eliminate / remove, anomalies ; 12 <i>ref. to</i> low risk ; or A named hazard and risk and precaution e.g. yeast and allergy and wear gloves / mask / goggles e.g. TTC and irritant and wear gloves / mask / goggles	7
1(c)(i)	<i>idea of</i> how close the (sample) <u>mean</u> , is to the, true / population, <u>mean</u> ;	1
1(c)(ii)	either 95% of, the / all / repeated, data, would be expected to lie within this range ; or At 1 / 3 / 4, hours, the (sample) <u>mean</u>, was reliable / AW, because the, confidence intervals / CI, do not overlap ;	1

Question	Answer	Marks
1(c)(iii)i	<i>from the graph / Fig. 1.3 / experiment 1</i> pH 6.0 gives the highest absorbance at 4 hours incubation ; or <i>from the table / Table 1.1 / experiment 2</i> at 30°C at pH 6 absorbance is highest ; <i>from the graph / Fig. 1.3 / experiment 1</i> the CI / (standard) error bars, (for pH 6), do not overlap (at 4 hours) ; or <i>from the table / Table 1.1 / experiment 2</i> standard errors / S_M, do not overlap (with other, temperatures / pHs) ;	2

Question	Answer	Marks
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